

**ADDIS ABABA UNIVERSITY COLLEGE OF HEALTH
SCIENCE SCHOOL OF PUBLIC HEALTH**

**BIostatISTICS FINAL EXAMINATION FOR GRADUATE
ANESTHESIA STUDENTS**

Exam date; April 28, 2016.

Time allowed: 2:30 hr

Name _____

ID no _____

▪ ***Instructions***

- *Write your answer in legible and neat way*
- *Write your answer in **Bold letters** for MCQ*
- Sharing calculator strictly forbidden use your own calculator
- Side talk is not allowed if you have any question ask examiner
- Switch off your mobile phone
- *Any attempt for cheating will result in disqualification of your result.*

Part one multiple choice

1. The sum of deviations of the individual data elements from their mean is sometimes greater than and sometimes less than zero, depending on the data elements.
A True B False
2. A measure of variability tells us how well the measure of central tendency represents the entire set of scores
A True B false
3. Which of the following is qualitative variables
A. Parity B. Diastolic blood pressure C. Diagnosis D. All of the above
4. Qualitative data can be graphically represented by using a(n)
A. histogram B. frequency polygon C. Ogive D. bar graph
5. Fifteen percent of the students in a school of public Health are track in reproductive Health, 20% Health service management, 30% in Health informatics, and 20% in field epidemiology, and 15% in epidemiology. The graphical device(s) which can be used to present these data is (are)
A. A line graph B. a Histogram C. A pie chart D. leaf and stem E. All
6. Which of the following is a measure of dispersion?
A. Percentiles B. quartiles C. inter-quartile range D. all of the above
7. The sample variance
A. is always smaller than the true value of the population variance
B. is always larger than the true value of the population variance
C. could be smaller, equal to, or larger than the true value of the population variance
D. can never be zero
8. which of the following diagrammatic representation used to interpret deviation from normality
A. Histogram, B. Bar chart C pie chart D. All E. none
9. Two events, A and B, are mutually exclusive and each has a nonzero probability. If event A is known to occur, the probability of the occurrence of event B is
A. One B. any positive value C. **zero** D. any value between 0 to 1
10. For a continuous random variable x, the probability density function $f(x)$ represents
A. the probability at a given value of x C. the area under the curve at x
B. the area under the curve to the right of x D. the height of the function at x
11. Central Limit Theorem refers to:
A. The tendency of the mean of sample means to equal the population mean.
B. The tendency of the mean of population means to equal the sample mean.
C. The tendency of the sample means to be different from the population means
D. All of the above is true
12. Under what circumstances should we be cautious about using the mean as a measure of central tendency?
A. When the data is skewed. B. When data is negatively skewed
C. When data is positively skewed. D. The entire above E. None
13. **The purpose** of descriptive statistics is to
A. Determine if the sample data accurately describe the population.
B. Simplify and organize large sets of data.
C. Help us decide whether population means are equal.
D. all of the above

14. If two events are mutually exclusive and both have probability > 0 , then
- A. They must also be independent.
 - B. They cannot also be independent.
 - C. They must also be complements.
 - D. They cannot also be complements.
15. Which of the following is correct about a simple random sample
- A. Best approach if investigator interested to address minority group
 - B. usually not requires that a sampling frame be constructed
 - C. usually results in a sample that is representative of the population from which the sample was selected
 - D. all of the above
16. Which of the following are true statements?
- A. Sampling techniques that use probability techniques effectively eliminate bias.
 - B. Voluntary response samples often under represent people with strong opinions.
 - C. Questionnaires with non neutral wording are likely to have response bias.
 - D. None
17. Which of the following are true statements about sampling error?
- A. Sampling error can be eliminated only if a survey is both extremely well designed and extremely well conducted.
 - B. Sampling error concerns natural variation between samples, is always present, and can be described using probability.
 - C. Sampling error is generally smaller when the sample size is larger.
 - D. All of the above
 - E. B and C are true
18. All of the followings are advantage of stratified sampling except
- A. Representativeness
 - B. Representation of the minority group
 - C. Having separate sampling frame
 - D. Probability selection of individuals in the strata
19. A Medical student plans a study to examine the depth of belief in God among the adult population. He obtains a simple random sample of 100 adults as they leave church one Sunday morning. All but one of them agrees to participate in the survey. Which of the following are true statements?
- A. Proper use of chance as evidenced by the simple random sample makes this a well designed survey.
 - B. The high response rate makes this a well-designed survey.
 - C. Selection bias makes this a poorly designed survey.
 - D. All of the above
 - E. None
20. Which of the following best describes estimate obtained from census?

- A. Both sampling and measurement error
 - B. Sampling errors but not measurement error
 - C. Measurement error but not sampling error
 - D. Neither sampling error nor measurement error
21. Which of the following true about standard normal distribution?
- A. distribution which it's mean and standard deviation is equal to one
 - B. distribution which it's mean and standard deviation can has any real value
 - C. distribution which it's mean can be determined by using a z-score
 - D. distributions which it's mean zero and standard deviation one.
 - E. all the above
22. Which of the following is used to measure strength of the association
- A. Chi-square test B. Fisher exact test C. Binomial test D Relative Risk Ratio
23. Assuming a sample is taken and its mean is calculated as 240. For a 90 percent confidence interval for μ , the upper limit comes to 280, what is the lower limit of this confidence interval?
- A. 220 B. 240 C. 200 D. cannot be determined from the information
24. For using a t distribution table, which of the following is a necessary condition?
- A. The population is large B. σ is not known but s is known C. n is small
 - D. (B) and (C) E. (A) and (B)
25. When a test of hypothesis involves the mean of a normal population with a known standard deviation, the comparison should be between:
- A. The observed value of $\bar{X}$ with the critical value of Z
 - B. The observed value of Z with the critical value of $\bar{X}$
 - C. The observed value of Z with the critical value of Z
 - D. The observed value of $\bar{X}$ with the critical value of $\bar{X}$
 - E. Either (c) or (d)
26. The margin of error can't be reduced if;
- A. the sample standard deviation is lower
 - B. The sample size is increased
 - C. The confidence level is decreased
 - D. The variance is higher
27. A result is called "statistically significant" whenever
- A. The null hypothesis is true.
 - B. The p-value is less than to the significance level.
 - C. If the confidence interval includes one D. All of the above.
28. What will happen if the researcher increases the level of Type I error without making any other changes?
- A. The level of Type II error will increase.
 - B. The level of Type II error will decrease.
 - C. Alpha will decrease.

- D. There will be no change in either alpha or the level of Type II error.
29. If two populations are different on their population means and the inferential statistical test leads you to conclude that the populations are equivalent, you have
- A. Made a Type I error.
 - B. Made a Type II error.
 - C. Drawn a correct inference.
 - D. Inflated your alpha
30. Which of the following sampling method used specially in hidden population
- A. Judgmental sampling
 - B. Simple random sampling
 - C. Snow ball sampling
 - D. All
31. Cholesterol levels are measured on a random sample of 5,000 persons, and the sample standard deviation is calculated. Suppose a second survey were repeated in the same population, but the sample size tripled to 15,000. Then which of the following is true?
- A. The new sample standard deviation would tend to be smaller than the first and approximately about one-third the size.
 - B. The new sample standard deviation would tend to be larger than the first and approximately about three times the size.
 - C. The new sample standard deviation would tend to be larger than the first, but we cannot approximate by how much
 - D. None of the above is true because there is no reason to believe one standard deviation would tend to be larger than the other.
32. Which of the following statement is true
- A. 95% confidence interval is narrower than a 99% confidence interval.
 - B. In simple linear regression residual is difference between observed and predicted value
 - C. Power is the probability of accepting false hypothesis
 - D. All of the above true
33. Convenience sampling is an example of
- A. probabilistic sampling
 - B. stratified sampling
 - C. non probabilistic sampling
 - D. cluster sampling
34. When a random sample is to be taken from a population and a statistic is to be computed, the statistic can also be thought of as
- A. A point estimate
 - B. A random variable
 - C. Both of the above
 - D. None of the above
35. Two coins are tossed simultaneously. If one of them turned head, what is the probability that the other one also turn head?
- A. 0.25
 - B. 1
 - C. 0.5
 - D 0.75
36. Null and alternative hypotheses can be statements about which of the following?
- A. Population parameters.
 - B. Sample estimates.
 - C. Sample statistics.
 - D. it depends - sometimes population parameters and sometimes sample statistics.
37. The confidence level for a confidence interval for a mean is
- A. The probability of making a Type 1 error if the interval is used to test a null hypothesis about the population mean.

- B. The probability that individuals in the population have values that falls into the interval.
 - C. The probability that the procedure provides an interval that covers the sample mean.
 - D. The probability that the procedure provides an interval that covers the population mean.
38. In chi-squared test for a 5 by 3 contingency table:
- A. Variable must be quantitative
 - B. Observed frequency are compared to expected frequency
 - C. There are 15 degree of freedom
 - D. At least 15 cells must have expected values greater than five
39. Which of the following probability sampling methods doesn't needs all of the population list under study.
- A. Systematic sampling
 - B. Simple random sampling
 - C. Cluster sampling
 - D. Interval sampling.
40. Which of the following statement is true
- A. Reporting confidence interval instead of p-value provide more information about precision of the study
 - B. The main drawback of a one-tailed test is that, required larger number of study subject as compared with two- tailed.
 - C. two-tail test is the loss of the ability to test for unanticipated results
 - D. All of the above true

Part two short answer (Do not calculate, just give final result)

1. A study was made of a sample of 25 records of patients seen at a chronic disease hospital on an outpatient basis. The mean number of outpatient visits per patient was 4.8, and the sample standard deviation was 2. Can it be concluded from these data that the population mean is greater than four visits per patient? Let the probability of committing a type I error be 05.
 - A. State null hypothesis_____
 - B. State alternative hypothesis_____
 - C. Which test statistics is the appropriate_____
 - D. What is point estimate for the above question_____
2. The mean serum-creatinine level measured in 12 patients 24 hours after they received a newly proposed antibiotic was 1.2 mg/dl. If the mean and standard deviation of serum creatinine in the general population are 1.0 and 0.4 mg/dl, respectively, then, using a significance level of .05.
 - A. Calculate degree of freedom and standard error_____
 - B. What test appropriate test statistics_____
 - C. What is 95% confidence interval based on given information_____
3. The following are result from a study on immunization (%) and infant mortality rate (per 1000 live birth in selected countries of the world
 - A. Results: r (coefficient of correlation) = -0.79 with 95% confidence interval of (-0.74,-0.84) and degrees of freedom of 78. What was the sample size for study_____.
 - B. Interpret the coefficient of correlation._____
 - C. Was the relationship significant? why_____
 - D. Assuming that regression equation for the two variable $y=226-2.1X$, as per give information, give interpretation of the coefficients?